Question Paper

Physics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If the intermolecular forces vanish away, the volume occupied by the molecules contained in 4.5 kg water at standard temperature and pressure will be given by

A. $5.6\ \mathrm{m}^3$

B. $4.5\ \mathrm{m}^3$

C. $11.2\ \mathrm{L}$

D. $11.2\ \mathrm{m}^3$

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A physical quantity P is related to four observables a, b, c and d as $P = \frac{\sqrt{ab} \cdot d^{\alpha}}{\sqrt{c}}$ (α is constant) The percentage errors in a, b, c and d are 0.5% in each. If the percentage error in P is 2%, then α is.....

A. $\frac{5}{2}$

B. $\frac{2}{5}$

C. $\frac{3}{4}$

D. $\frac{3}{2}$

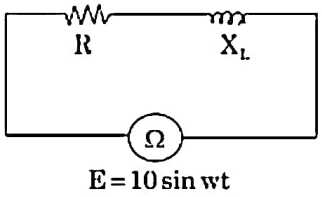
@shubham_sharma715

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

An AC circuit having supply voltage E consists of a resistor of resistance 3Ω and an inductor of reactance 4Ω as shown in the figure. The voltage across the inductor at $t = \frac{T}{2}$ is



A. 2 V

B. 10 V

C. Zero

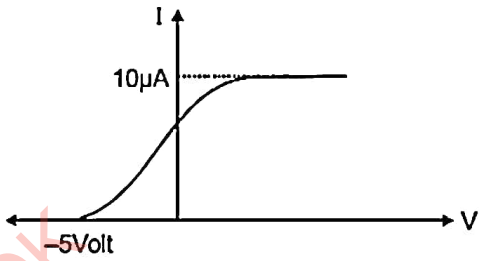
D. 4.8 V

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In the photoelectric experiment, if we use a monochromatic light, the $I - V$ curve is as shown. If work function of the metal is $2eV$, estimate the power of light used. (Assume efficiency of photo emission $= 10^{-3}\%$, i.e. number of photoelectrons emitted are $10^{-3}\%$ of number of photons incident on metal.)



A. 2 W

B. 5 W

C. 7 W

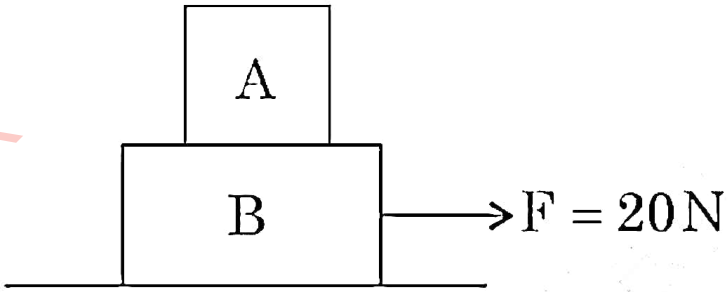
D. 10 W

Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Block A of mass 2 kg is placed over block B of mass 8 kg. The combination is placed over a smooth horizontal surface. Coefficient of friction between A and B is 0.5. A horizontal force of 20 N is applied on block B. The force of friction on A is



A. 10 N

B. 8 N

C. 6 N

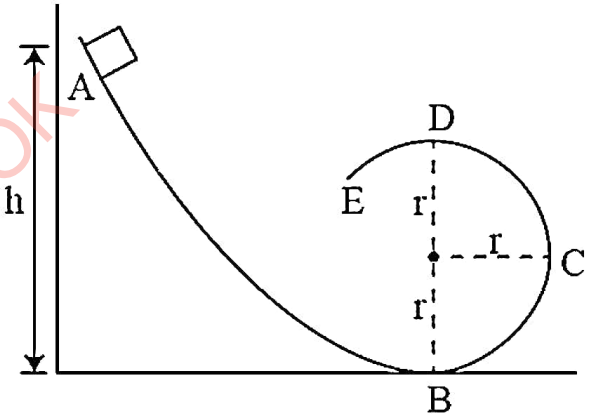
D. 4 N

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

ABCDE is a channel in the vertical plane, part $BCDE$ being circular with radius r. A block is released from A and slides without friction and without rolling. The block will complete the loop if h is :-



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A. $h \leq \frac{3}{2} r$

B. $h \geq \frac{5}{2} r$

C. $h \geq \frac{3}{2} r$

D. $h \leq \frac{5}{2} r$

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The image formed by a convex mirror of focal length 30 cm is a quarter of the size of the object. The distance of the object from the mirror is----

A. 30 cm

B. 90 cm

C. 120 cm

D. 60 cm

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A proton and an electron enter into a magnetic field $\mathbf{B}$ perpendicular to it with speed v and $10v$. Then, the ratio of the radii of their circular paths is (given that, proton is 1820 times heavier than electron)____

A. 1820 : 1

B. 18200: 1

C. 1: 182

D. 182 : 1

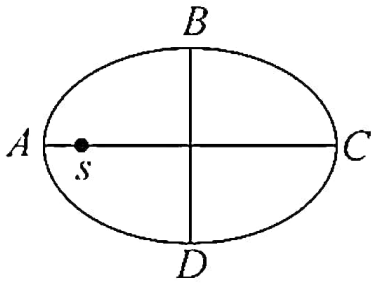
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Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A planet is revolving around the sun. Which of the following is correct statement?



A. The time taken in travelling DAB is less than that for BCD

B. The time taken in travelling DAB is greater than that for BCD

C. The time taken in travelling CDA is less than that for ABC

D. The time taken in travelling CDA is lgreater than that for ABC

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

For a Young's double slit experiment by coherent light source the intensity at central maxima was found to be I_0. The intensity of light formed at a distance of $\beta/3$ from central maxima is (where ' β ' is the fringe width)....

A. $\frac{I_0}{4}$

B. $\frac{3I_0}{4}$

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C. $\frac{I_0}{2}$

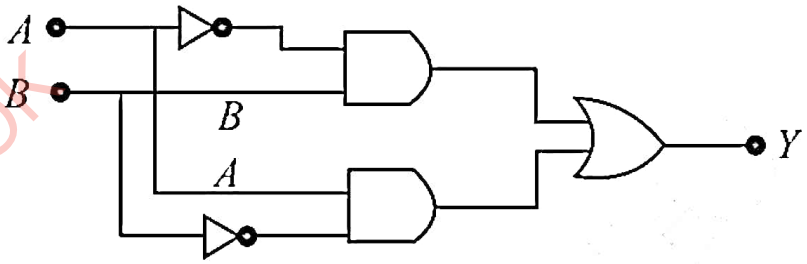
D. $\frac{I_0\sqrt{3}}{2}$

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The truth table for the following logic circuit is



A.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

B.

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	1

C.

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

D.

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	1

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Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A big water drop is formed by the combination of n small water drops of equal radii. The ratio of the surface energy of n drops to the surface energy of the big drop is

A. $n^2 : 1$

B. $n : 1$

C. $\sqrt{n} : 1$

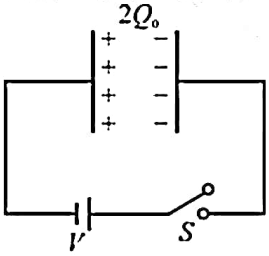
D. $\sqrt[3]{n} : 1$

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A capacitor of capacitance C having initial charge $2Q_0$, is connected to a battery of potential difference $V = \frac{Q_0}{C}$ as shown, then workdone by the battery is



A. $\frac{Q_0^2}{2C}$

B. $\frac{3Q_0^2}{2C}$

C. $\frac{3Q_0^2}{C}$

D. $\frac{2Q_0^2}{3C}$

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Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

One end of wire $2\ \mathrm{m}$ long and diameter $2\ \mathrm{mm}$ is fixed in a ceiling. A naughty boy of mass $10\ \mathrm{kg}$ jumps to catch the free end and stays there. The change in length of wire is [Take $g = 10\ \mathrm{m\,s^{-2}},\ Y = 2 \times 10^{11}\ \mathrm{N\,m^{-2}}$]

A. $31.85 \times 10^{-5}\ \mathrm{m}$

B. $2\ \mathrm{mm}$

C. $3\ \mathrm{mm}$

D. $4\ \mathrm{m}$

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The equation that represents magnetic field of a plane electromagnetic wave which is propagating along x-direction with wavelength $10\ \mathrm{mm}$ and maximum electric field $60\mathrm{Vm^{-1}}$ in y-direction is (where, $c =$ speed of light)

A. $\left(6 \times 10^{-7}\right) \sin[0.2\pi(ct - x)]\hat{k}T$

B. $\left(2 \times 10^{-7}\right) \sin[200\pi(ct - x)]\hat{k}T$

C. $\left(2 \times 10^{-7}\right) \sin[200\pi(ct - x)]\hat{i}\ \mathrm{T}$

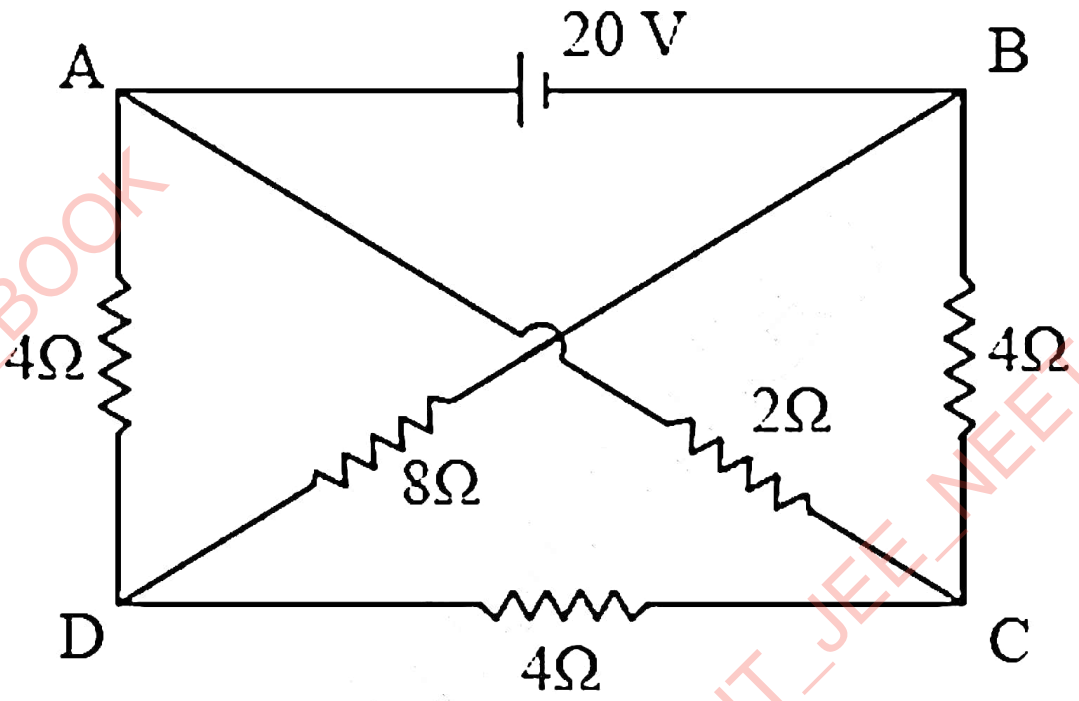
D. $\left(6 \times 10^{-7}\right) \sin[20\pi(ct - x)]\hat{i}T$

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider the circuit shown in the figure.



Which of the following statements are correct?

A. Current through the battery is 5 A

B. Current through branch AD is 2 A

C. Current through branch BC is 3 A

D. The potential difference between points D and C is 4 V

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Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The position vector of a particle is given as $\vec{r} = (t^2 - 4t + 6)\hat{i} + (t^2)\hat{j}$. The time after which the velocity vector and acceleration vector becomes perpendicular to each other is equal to

A. 1sec

B. 2sec

C. 1.5sec

D. not possible

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Two identical containers A and B having the same volume of an ideal gas at the same temperature have the mass of the gas as m_1 and m_2 respectively and $2m_1 = 3m_2$. The gas in each cylinder expands isothermally to double of its volume. If the change in pressure in A is 300 Pa, then the change in pressure in B is

A. 200 Pa

B. 300 Pa

C. 400 Pa

D. 500 Pa

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Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A sample of hydrogen like atoms produces an emission spectrum consisting of 10 wavelength arising out of all transitions possible. During this process the maximum angular impulse on an electron going from higher energy level to a lower energy level is :-

A. $\frac{h}{2\pi}$

B. $\frac{h}{\pi}$

C. $\frac{2h}{\pi}$

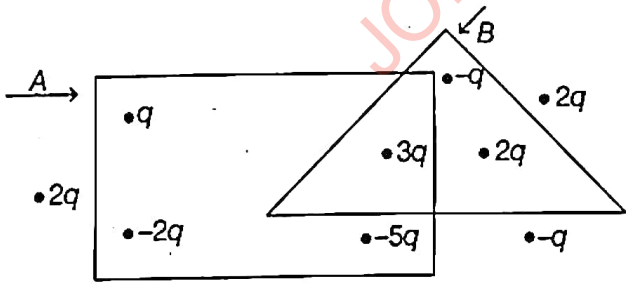
D. $\frac{3h}{2\pi}$

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The distribution of some charges on two Gaussian surfaces A and B are as shown in the figure. If ϕ_A and ϕ_B are electric fluxes linked with the surfaces A and B respectively. then $\frac{\phi_A}{\phi_B} =$



A. $-\frac{1}{5}$

B. -3

C. $-\frac{3}{2}$

D. $-\frac{3}{4}$

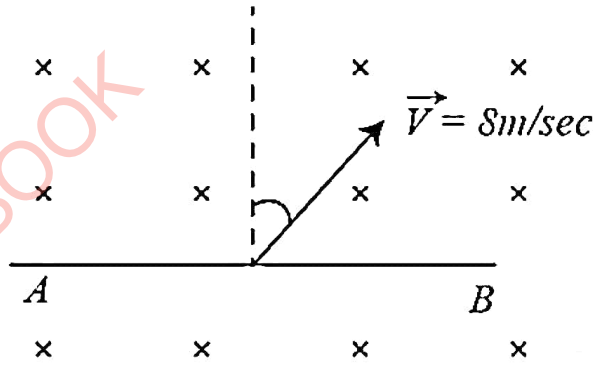
Physics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

A metal rod AB of length 50 cm is moving at a velocity 8 ms^{-1} in a magnetic field of 2 T. If the field is at 60° with the plane of motion as shown in the figure, then the potentials V_A and V_B are related by $V_A - V_B = K$, find K.



Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

A central fringe of the interference produced by light of wavelength 6000Å is shifted to the position of 5th bright fringes by introducing a thin glass plate of refractive index 1.5. Calculate the thickness of the plate. If it is $n \times 10^{-4}$ cm. Find n.

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

When a body of mass 1.0 kg is suspended from a certain light spring hanging vertically, its length increases by 5 cm. By suspending 2.0 kg block to the spring and if the block is pulled through 10 cm and released, the maximum velocity in it in m/s is : (Acceleration due to gravity $= 10$ m/s^2)

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

All the edges of a block with parallel faces are unequal. Its largest edge is twice its shortest edge. The ratio of the maximum to the minimum resistance between parallel faces is

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

A particle of mass $15\ \text{kg}$ is moving with a uniform speed $8\ \text{ms}^{-1}$ in xy-plane along the line $3y = 4x + 10$, then the magnitude of its angular momentum about the origin in $\text{kg} \cdot \text{m}^2\ \text{s}^{-1}$ is ... $\left(\sin 53^\circ = \frac{4}{5}\right)$

Chemistry Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

An indicator has a pK_a of 7.5. What is the ratio of the concentration of the base form to the concentration of the acid form ($[A^-]/[HA]$) in a buffer solution in which the pH is 6.5 ?

A. 10/1

B. 1/1

C. 1/10

D. 6.5/7.5

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Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Match the following

	List - I		List - II
a)	Vitamin E	i)	Meat
b)	Vitamin K	ii)	Sunflower oil
c)	Vitamin B_{12}	iii)	Egg white
d)	Vitamin B_2	iv)	Exposure to sunlight
		v)	Green leafy vegetables

The correct answer is

A. (a) iii (b) ii (c) iv (d) i

B. (a) ii (b) v (c) i (d) iii

C. (a) i (b) iii (c) ii (d) v

D. (a) iv (b) i (c) v (d) ii

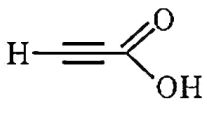
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Question No. 3

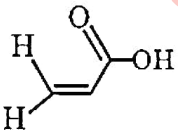
Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

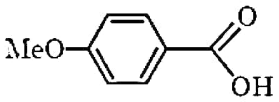
The correct order of acid strength of the following carboxylic acids is -



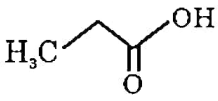
I



II



III



IV

A. $I > III > II > IV$

B. $III > II > I > IV$

C. $II > I > IV > III$

D. $I > II > III > IV$

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The IUPAC name of the following complex is $\left[Cr(NH_3)_3(H_2O)_2Cl\right]Cl_2$.

A. Triamminediaqua chlorido chromium (III) ch!oride.

B. Diaquatriammine chlorido chromium (III) chloride.

C. Chlorido diaquatriammine chromium (III) chloride.

D. Triammine diaqua trichlorido chromium (III).

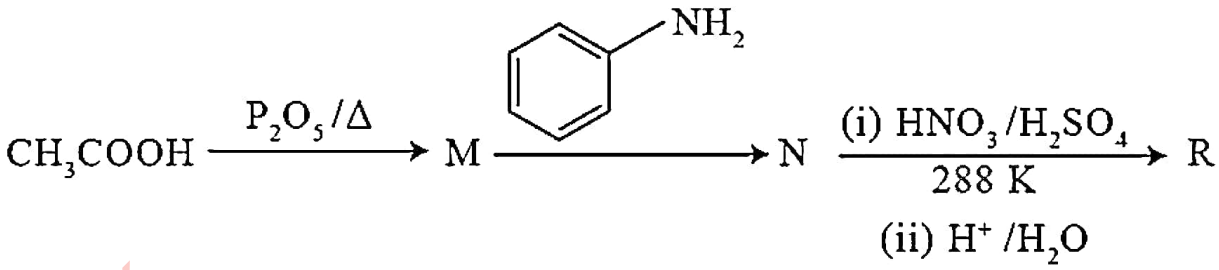
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Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

What is the major product 'R' in the following reaction sequence?



A. o-Nitroaniline

B. m-Nitroaniline

C. p-Nitroaniline

D. p-Aminobenzene sulphonic acid

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Molarity of a 50 mL H_2SO_4 solution is 10.0M. If the density of the solution is 1.4 g/cc, calculate its molality (mark answer to nearest integer)--

A. 12

B. 24

C. 72

D. 48

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Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A substance A , which sublimes on heating, forms nitrate when treated with HCl and a small quantity of $NaNO_2$. Suggest from this information a suitable test for A.

A. Nessler's test

B. Decolorization with acidified $KMnO_4$

C. Brown ring test

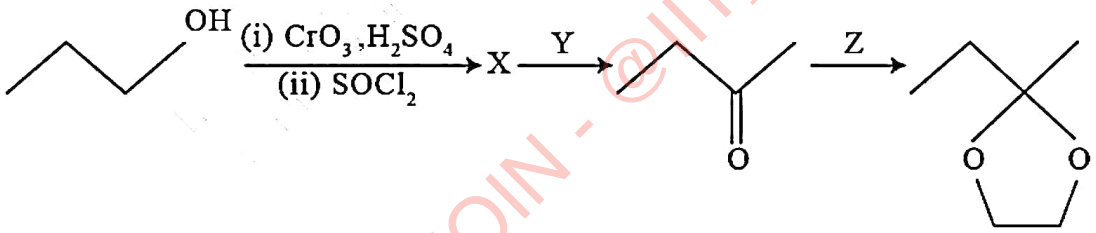
D. Lime water test

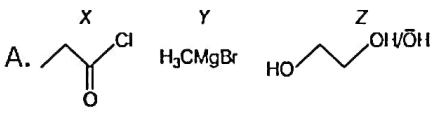
Question No. 8

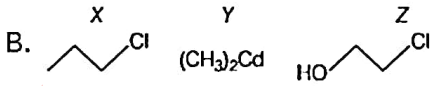
Only One Option Correct Type

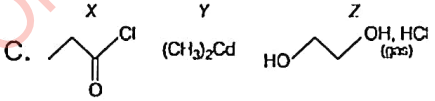
Each question has multiple options out of which ONLY ONE is correct.

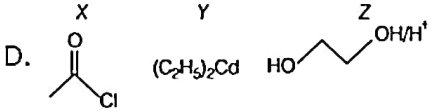
Identify X, Y, Z in the following reaction sequence



A. 

B. 

C. 

D. 

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Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Match the following.

	Column I		Column II
A.	Acetaldehyde, vinyl alcohol	1.	Enantiom
B.	Eclipsed and staggered ethane	2.	Tautomer
C.	(+)2-butanol, (-)2-butanol	3.	Chain isor
D.	Methyl-n-propylamine and diethylamine	4.	Conforma
		5.	Metamers

A, B, C, D match with

A. 1 4 3 5

B. 2 4 1 5

C. 5 1 4 2

D. 5 1 3 2

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Column I	Column II
A. He	i. Highest electron gain enthalpy
B. Cl	ii. Most electropositive element
C. Cs	iii. Strongest reducing agent
D. Li	iv. Highest ionisation energy

The correct match of the contents in column I with those in column II is

:

A. A-(iii), B-(i), C-(ii), D-(iv)

B. A-(iv), B-(iii), C-(ii), D-(i)

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C. A-(i), B-(ii), C-(iii), D-(iv)

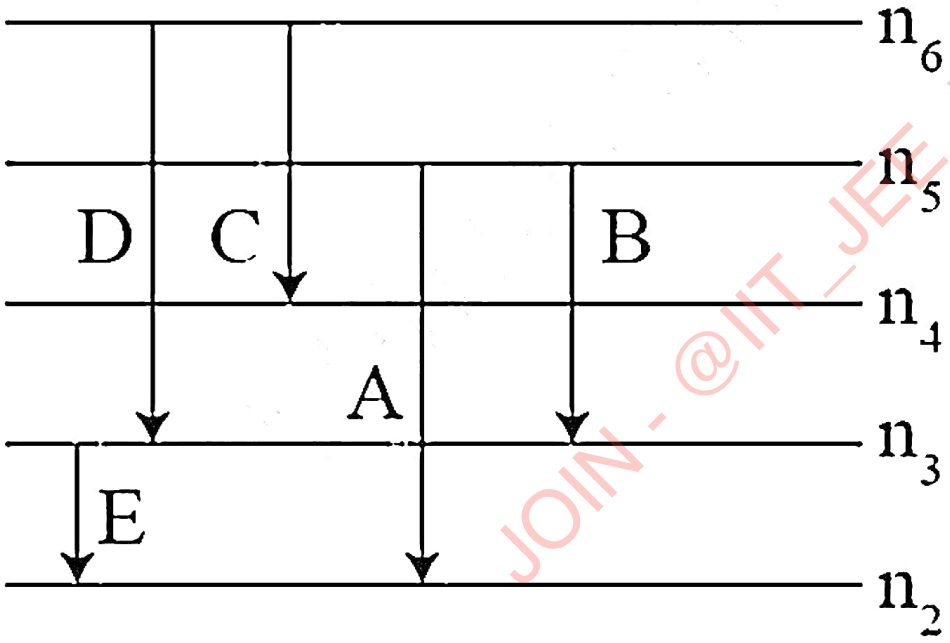
D. A-(iv), B-(i), C-(ii), D-(iii)

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

For a hypothetical H-like atom, which follows Bohr model, some spectral lines are observed as shown. If it is known that line E belong to visible region., then the line, which possibly belong to U.V. region is/are? (n_1 is not necessarily the ground state)



A. B and D

B. D only

C. C only

D. A only

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Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Oxygen is more electronegative than sulphur. Yet H_2S is acidic while H_2O is neutral. This is because-

A. Water is a highly associated compound

B. Molecular mass of H_2S is more than that of H_2O

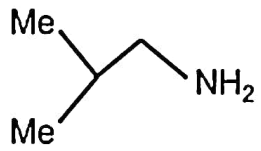
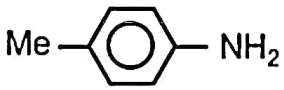
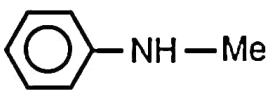
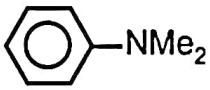
C. H_2S is gaseous under ordinary conditions while H_2O is a liquid

D. $H-S$ bond is weaker than $H-O$ bond

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

	Column I		Column II
S. NO.	Compound		Characteristics
a.		p.	Liebermann's nitroso reaction
b.		q.	Evolution of N_2 with HNO_2
c.		r.	Dye test
d.		s.	Green colour with HNO_2
		t.	Carbylamine test

A. $(a \rightarrow q, t; b \rightarrow r, t; c \rightarrow p; d \rightarrow s)$

B. $(d \rightarrow q, r; c \rightarrow t, s; b \rightarrow q; a \rightarrow r)$

C. $(a \rightarrow q, t; b \rightarrow r, s; c \rightarrow p; d \rightarrow s)$

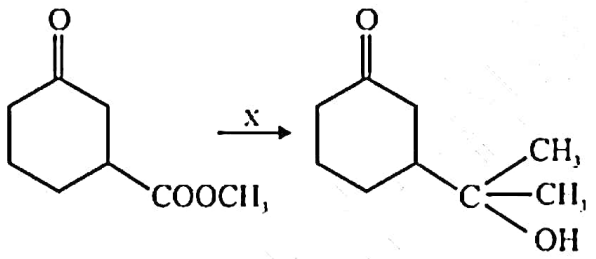
D. $(a \rightarrow p, q; b \rightarrow s, t; c \rightarrow p; d \rightarrow t)$

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider the following reaction



The reagent ' X in the reaction is----

A. $HOCH_2 - CH_2OH|H^+, LiAlH_4|$ ether, $2CH_3MgBr, H_3O^+$

B. $2CH_3MgBr$ and H_3O^+

C. $HOCH_2 - CH_2OH|H^+, 2CH_3MgBr, H_3O^+$

D. $HOCH_2 - CH_2OH - |H^+, H_2|Pt, CH_3OH, H^+$

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which one of the following lanthanide ions does not exhibit paramagnetism?

A. Lu^{3+}

B. Ce^{3+}

C. Eu^{3+}

D. Yb^{3+}

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Steam distillation process cannot be used for purifying which of the following?

A. Aniline

B. p-nitrophenol

C. Toluene

D. Nitrobenzene

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Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Select the set having incorrect statements given here.

(i) Manganese exhibits $+7$ oxidation state

(ii) Zinc forms coloured ions

(iii) $[CoF_6]^{3-}$ is diamagnetic

(iv) Sc forms $+4$ oxidation state

(v) Zn exhibits only $+2$ oxidation state

A. (i), (ii) and (iii)

B. (ii), (iii) and (iv)

C. (i), (iii) and (iv)

D. (i), (iii) and (v)

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The sublimation energy of $I_2(s)$ is $57.3\ kJ\ mol^{-1}$ and the enthalpy of fusion is $15.5\ kJ\ mol^{-1}$. The enthalpy of vaporization of I_2 is-

A. $41.8\ kJ\ mol^{-1}$

B. $-41.8\ kJ\ mol^{-1}$

C. $72.8\ kJ\ mol^{-1}$

D. $-72.8\ kJ\ mol^{-1}$

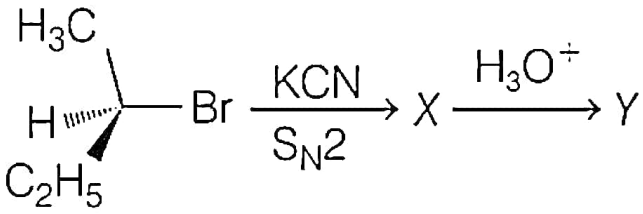
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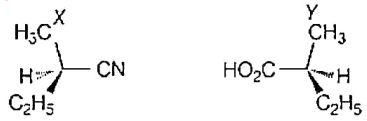
Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

What are X and Y in the following reaction sequence?



A. 

B. 

C. 

D. 

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A solution when treated with dimethyl glyoxime gives a rosy red complex. The metal present is:

A. Ni

B. Al

C. Co

D. Mn

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Chemistry Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

At 400 K, in a 1.0 L vessel, N_2O_4 is allowed to attain equilibrium, $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ At equilibrium, the total pressure is 600 mmHg, when 20% of N_2O_4 is dissociated. The value of K_n for the reaction is

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

The bond order of O_2^+, O_2, O_2^- and O_2^{2-} will be A, B, C, D respectively, Find A + 2 B + C − D.

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

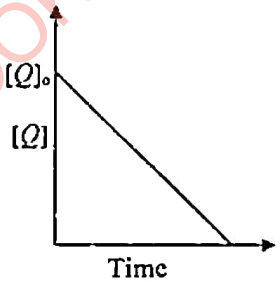
Consider a titration of potassium dichromate solution with acidified Mohr's salt solution using diphenyl amine as indicator. The number of moles of Mohr's salt required per mole of dichromate is

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

In the reaction, $P + Q \rightarrow R + S$
The time taken for 75% reaction of P is twice the time taken for 50% reaction of P. The concentration of Q varies with reaction time as shown in the figure. The overall order of the reaction is



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Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

The molar conductivity of a solution of a weak acid HX $(0.01\mathrm{M})$ is 10 times smaller than the molar conductivity of a solution of a weak acid $\mathrm{HY}(0.10\mathrm{M})$. If $\lambda^0_{X^-} \approx \lambda^0_{Y^-}$, the difference in their $\mathrm{pK_a}$ values, $\mathrm{pK_a(HX) - pK_a(HY)}$, is (Consider degree of ionization of both acids to $<< 1$)

Mathematics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If B, C are square matrices of same order such that $C^2 = BC - CB$ and $B^2 = -I$, where I is an identity matrix, then the inverse of matrix $(C - B)$ is

A. C

B. $C + B$

C. $C - B$

D. I

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

3 women and 15 men are to be arranged in a row such that there should be atleast 2 men between all the two consecutive women. Then number of such arrangements is :

A. $^{14}\mathrm{C}_4 3\,!$

B. $3!\ 15!$

C. $^{14}\mathrm{C}_{12}$

D. $^{14}\mathrm{C}_3\ 3!\ 15!$

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Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let M be the foot of the perpendicular from a point P on the parabola $y^2 = 8(x-3)$ onto its directrix and let S be the focus of the parabola. If $\triangle SPM$ is an equilateral triangle, then P is equal to

A. $(4\sqrt{3}, 8)$

B. $(8, 4\sqrt{3})$

C. $(9, 4\sqrt{3})$

D. $(4\sqrt{3}, 9)$

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The minimum value of the function, $f(x) = x^{3/2} + x^{-3/2} - 4\left(x + \frac{1}{x}\right)$. for all permissible real values of x, is

A. -10

B. -6

C. -7

D. -8

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Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A data consists of n observations: $x_1, x_2, \ldots, x_n$. If $\sum_{i=1}^{n}(x_i+1)^2 = 11n$ and $\sum_{i=1}^{n}(x_i-1)^2 = 7n$, then the variance of this data is

A. 5

B. 8

C. 6

D. 7

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $(1!)^2 + (2!)^2 + (3!)^2 + \ldots + (99!)^2 + (100!)^2$ is divided by 100 , then the remainder obtained is

A. 27

B. 28

C. 17

D. 14

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Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

For some real number λ, if the area of the triangle having $\vec{a} = 3\hat{i} - \hat{j} + \lambda\hat{k}$ and $\vec{b} = \lambda\hat{i} + \hat{j} - 3\hat{k}$ as two of its sides is $\frac{\sqrt{195}}{2}$, then the number of distinct possible values of λ is

A. 4

B. 3

C. 2

D. 1

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The number of points (x, y) satisfying $x^2 + y^2 \leq 4$ & $\tan^4 x + \cot^4 x + 1 = 3\sin^2 y$ is equal to

A. 4

B. 2

C. 3

D. 5

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Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

For any three positive real numbers a, b and $c, 4\left(9c^2+b^2\right)+9\left(a^2-2ac\right)=6b(2c+a)$. Then:

A. b, c and a are in A.P.

B. c, a and b are in A.P.

C. a, b and c are in G.P.

D. b, c and a are in G.P.

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $\int \frac{x^2\left(x\sec^2 x+\tan x\right)}{(x\tan x+1)^2}\,dx = A\log$ $\left(|x\sin x+\cos x|\right)+B\,\frac{f(x)}{(x\tan x+1)}+C$, then $f(A+B)=$

A. 1

B. 0

C. -1

D. 2

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Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The number of points $P(x, y)$ with natural numbers as coordinates that lie inside the quadrilateral formed by the lines $2x + y = 2, x = 0, y = 0$ and $x + y = 5$ is

A. 12

B. 10

C. 6

D. 4

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The solution of the differential equation $xy \log\left(\frac{x}{y}\right) dx + \left\{y^2 - x^2 \log\left(\frac{x}{y}\right)\right\} dy = 0$ is

A. $\frac{x^2}{2y^2} \log\left(\frac{x}{y}\right) - \frac{x^2}{4y^2} = -\log y + \log c$

B. $\frac{x^2}{2y^2} \log\left(\frac{x}{4y}\right) + \frac{x^2}{4y^2} = \log y + \log c$

C. $\log\left(\frac{x}{y}\right) - \frac{x^2}{4y^2} + \frac{x^2}{2y^2} \log y + \log c = 0$

D. none of these

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Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

There are n sets of observations given as $(1),\ (2\ ,3),\ (4\ ,5,6),\ (7\ ,8,9\ ,10),\ldots..$ The mean of the 13th set of observations is equal to

A. 70

B. 80

C. 75

D. 85

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If L_1 is a line passing through the point $\mathrm{P}(4,-3)$ and perpendicular to the line $3x-4y+k=0$, then the distance of P from the line $5x-3y-2=0$ measured along the line L_1 is

A. 5

B. $\sqrt{13}$

C. $\sqrt{41}$

D. 13

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Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $f(x) = (2k+1)x - 3 - ke^{-x} + 2e^{x}$ is monotonically increasing for all $x \in R$, then the least value of k is

A. 1

B. 0

C. $-\frac{1}{2}$

D. -1

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $\alpha = \sin\left(\cot^{-1}\left(\tan\left(\cos^{-1}\frac{2}{3}\right)\right)\right)$ and $\beta = \sin\left(\text{cosec}^{-1}\left(\cot\left(\tan^{-1}\frac{1}{3}\right)\right)\right)$ are the roots of the quadratic equation $ax^2 + bx + c = 0$ where a, b, c are integers and c is prime then the value of $(a+b+c)$ equals

A. 20

B. 11

C. 7

D. 2

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Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $f : R \to R$ be a function such that $f\left(\frac{x+y}{3}\right) = \frac{f(x)+f(y)}{3}$, $f(0) = 0$ and $f'(0) = 5$, then

A. $f(x)$ is a quadratic function

B. $f(x)$ is continuous but not differentiable

C. $f(x)$ is differentiable in R

D. $f(x)$ is bounded in R

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let R is a relation defined as $R = \{(1, 2), (2, 3), (3, 4)\}$. The minimum number of ordered pairs which should be added to make relation 'R' equivalence relation, are

A. 7

B. 9

C. 13

D. 11

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Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Two numbers 'a' and 'b' are chosen at random from the numbers 1, 2, 3,....30. The chance that $a^2 - b^2$ is divisible by '3' is

A. $\frac{9}{87}$

B. $\frac{12}{87}$

C. $\frac{15}{87}$

D. $\frac{47}{87}$

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If a and c are unit vectors and $|\vec{b}| = \sqrt{5}$. The angle between $\vec{a}$ and $\vec{c}$ is $\frac{\pi}{3}$. and $\vec{b} - 2\vec{c} = \alpha\vec{a}$ Then, α is a root of the equation:

A. $x^2 + 2x + 1 = 0$

B. $x^2 + x + 1 = 0$

C. $x^2 + x + 12 = 0$

D. $x^2 + 2x - 1 = 0$

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Mathematics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

If $\int_0^{\pi} \frac{\sin x(\sin x+1)e^{\sin x+\cos x}}{e^{\cos x}+1}\,dx = a + b\int_0^{\pi} e^{\sin x}\,dx$ where a and b are positive rational numbers, then find the value of $100\left(a^2+b^2\right)$.

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

Consider the system of equations
$ax + y + bz = 0,\ bx + y + az = 0$ and $ax + by + abz = 0$ where $a, b \in \{0,1,2,3,4\}$. The number of ordered pairs (a, b) for which the system has non-trivial solutions is

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

The area (in square units) bounded by the curves $y = 2x^2$ and $y = \max\{x - [x], x + |x|\}$ in between the lines $x = 0$ and $x = 2$ is

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Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

Let $L_1 : \frac{x-1}{1} = \frac{y-1}{2} = \frac{z-1}{k}$ and $L_2 : \frac{x-3}{1} = \frac{y-2}{-1} = \frac{z-3}{-1}$ be two intersecting lines and L_3 be a straight line which is perpendicular to L_1 and L_2 and passing through their point of intersection. If length of perpendicular from $(3, 1, 2)$ to the line L_3 is λ then $[\lambda]$ equals [Note: [y] denotes the greatest integer function less than or equal to y.]

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

For any integer k, let $\alpha_k = \cos\left(\frac{k\pi}{7}\right) + i\sin\left(\frac{k\pi}{7}\right)$, where $i = \sqrt{-1}$. The value of the expression $\frac{\sum_{k=1}^{12}|\alpha_{k-1}-\alpha_k|}{\sum_{k=1}^{3}|\alpha_{4k-1}-\alpha_{4k-2}|}$ is

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